

Soil Processes and Chemical Transport

Measurement and Characterization of Macropores by using AUTOCAD and Automatic Image Analysis

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ABSTRACT

Macropores (with equivalent diameters greater than 1600 μm) were quantified under no-till and conventional tillage practices by using AUTOCAD and automatic image analysis. To quantify macropores, undisturbed soil blocks (50 by 50 cm) were impregnated with plaster of paris slurry. After the plaster of paris set, soil layers were taken out in 5-cm increments to a total depth of 60 cm. At each soil plane pictures were taken by a 35-mm camera and macropores were traced on transparent acetate sheets. The pictures and transparent sheets were analyzed by using an automated image analyzer and AUTOCAD, respectively, for total number, perimeter, area, and size frequency distribution of macropores. The AUTOCAD method proved to be simpler and better than the image analysis technique. Larger and continuous cracks accounted for greater macropore area in conventional tillage sites than in no-till sites. On the other hand, more worm holes and root holes accounted for a greater number and larger perimeters of macropores in no-till sites in comparison to conventional tillage sites.

IT HAS BEEN well documented in the literature that macropores can cause rapid hydrologic transport of solutes in both saturated and unsaturated soils (Beven and Germann, 1982; Thomas and Phillips, 1979; White, 1985). Preferential transport of chemicals may contribute greatly to groundwater contamination because some of the nutrients and chemicals applied on the soil surface move through the macropores instead of moving through the soil matrix where they could be adsorbed, chemically degraded, or biologically transformed.

Beven and Germann (1982) defined macropores as the pores that permit nonequilibrium channeling flow, irrespective of their size. Beven and Germann (1982) have summarized the values of minimum equivalent diameter for the macropores used by various research workers. These values range from 30 μm to 3000 μm .

In general, two approaches have been recognized to quantify macropores (Warner et al., 1989). The first approach attempts to determine the influence of macropores by measuring flow characteristics, for example, plotting breakthrough curves (Anderson and Bouma, 1977a,b; Bouma and Wösten, 1979), determining saturated or unsaturated hydraulic conductivity (Germann and Beven, 1981a,b; Bouma et al., 1977), and using tension infiltrometers (Everts and Kanwar, 1989, 1990; Watson and Luxmoore, 1986; Moore et al., 1986).

The second approach is a direct approach that uses methods to determine the actual number, size, area distribution, and continuity of macropores for various field conditions. A direct approach is useful in developing mathematical models to predict flow of water and chemicals through the soil profile. These models use existing flow theory for soil matrix. The effect of macropores is incorporated in terms of physical boundary conditions for the flow model derived from soil morphological observations. For example, Bouma and Wösten (1984) estimated the volume of infiltrated water in a soil block with the help of morphological data. The model of Hoogmoed and Bouma (1980) also uses morphological data as input parameters. Yeh and Luxmoore (1982) developed a model which handles the input of specific macropore conditions or a statistical average summary. Beven and Germann (1981) have discussed the mathematics of modeling water flow through individual soil cracks.

Direct methods use thin sections or polished blocks of resin impregnated soil cores (Murphy et al., 1977a,b; Bullock and Murphy, 1980; Ringrose-Voase and Bullock, 1984), photographs of soil profiles in the field (Edwards et al., 1988; Lauren et al., 1988; Smettem and Collis-George, 1985; FitzPatrick et al., 1985; Mackie, 1987), or nondestructive methods such as computed tomography (or CT) scanning (Warner et al., 1989). In many cases dyes such as methylene blue, lissamine yellow, or rhodamine B have been used to trace the macropore network in a soil profile (Omoti and Wild, 1979; Smettem and Trudgill, 1983) and calculate functional volume of macropores (Bouma and Dekker, 1978).

This paper presents the results of a study carried out in a loamy soil to characterize macropores quantitatively within the upper 60 cm of the soil profile under no-till and conventional tillage systems. This study used the direct approach of macropore quantification, and the pores larger than 1600 μm were considered as macropores. Two methods, namely AUTOCAD and automatic image analysis, were used to quantify macropores. The results obtained from these two methods were compared. Tillage-induced effects on macroporosity also were investigated in this study.

MATERIALS AND METHODS

Experiments for the direct measurement of macropores in the field were conducted during the summers of 1988 and 1989 (June, July, and August) at the Agronomy and Agricultural Engineering Research Center in Boone County, IA. The predominant soil at the experimental site was a Nicollet loam, a fine-loamy, mixed, mesic Aquic Hapludoll in the Clarion-Nicollet-Webster soil association, derived from glacial till with a slope of less than 3%. The study area had been

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Table 1. Physical properties of the Nicollet soil at the experimental site (Kanwar et al., 1988).†

Depth cm	Particle size distribution			Organic matter	Porosity	Bulk density g/cm ³
	Sand	Silt	Clay			
	%					
0-15	42.0	35.2	22.8	4.3	44.0	1.49
15-30	35.7	38.2	26.1	4.0	49.0	1.36
30-45	34.1	38.4	27.5	3.2	51.0	1.30
45-90	38.0	36.0	26.0	2.6	49.0	1.37

† The values given are the average values for no-till and conventional till plots.

planted to continuous corn (*Zea mays* L.) under no-till and conventional tillage systems for the past 6 yr (Kanwar et al., 1988). Conventional tillage plots were moldboard plowed in the fall to a depth of 15 to 20 cm. Selected physical properties of the soil at the experimental site are listed in Table 1 (Kanwar et al., 1988).

Three areas of 50 by 50 cm with visible cracks were randomly selected in both the no-till and conventional tillage plots. Each 50 by 50-cm area was then cleaned with a brush to remove crop residues from the surface. A square metal frame (50 by 50 by 15 cm) was placed carefully over the area so that it enclosed a part of the network of cracks. Then the frame was pushed into the soil, leaving about 5 cm of the frame above the soil surface. This served to define the sample area and to economize the amount of plaster of paris that was later poured on the soil surface. A mixture of plaster of paris and water was prepared (1:6 by volume) in small buckets and poured slowly enough to prevent ponding on the surface. Plaster of paris was poured into the visible cracks until the cracks were filled to the soil surface. The surface then was flooded with the mixture. The amount of plaster of paris required to fill the pores was different at replicate sites within the same tillage system. It varied from 8000 cm³ to 80 000 cm³. FitzPatrick et al. (1985) observed an even greater degree of variability with the amount of added plaster of paris in their study varying from 10 000 cm³ to 288 000 cm³.

After the plaster of paris hardened (in about 4 h) a 60-cm-wide trench was dug around the 50 by 50-cm soil column and a metal frame (50 by 50 by 5 cm) was clamped around the column to keep it from falling. Then the column soil was removed in 5-cm layers, to a total depth of 60 cm, by using spatulas and sharp knives. On some sites, after plaster of paris was added, the sites were left for about 1 wk without excavation, covered with a plastic sheet. No quantifiable difference was observed between the sites excavated after 1 wk and those excavated 4 h after pouring the plaster of paris mixture.

A transparent acetate sheet (50 by 50 cm) was spread on the surface of each plane (5-cm-depth intervals) and outlines of the plaster-filled cracks as well as unfilled worm holes, root holes, etc., were drawn with a permanent marker. These tracings were analyzed on a microcomputer by using a digitizing tablet and AUTOCAD — a computer-aided drafting package (Version 5.0), for number, perimeter, and areal extent of macropores. All the macropores mapped on transparent sheets were digitized but only the macropores of more than 1600- μ m-equivalent diameter were included in calculations of total area, number, and perimeter.

In addition to tracing the macropores on transparent sheets, a 35-mm camera was fixed on a tripod directly above the horizontal surface of the experimental block, and pictures of each plane were taken (at 5-cm intervals). A black-and-white film Panatomic-X (Kodak Co.) (ASA 32) was used to take the pictures. Different apertures and shutter speeds were tried at each depth so that the pictures having the best contrast could be selected for image analysis. Usually the pictures that were slightly underexposed (one F-stop less ex-

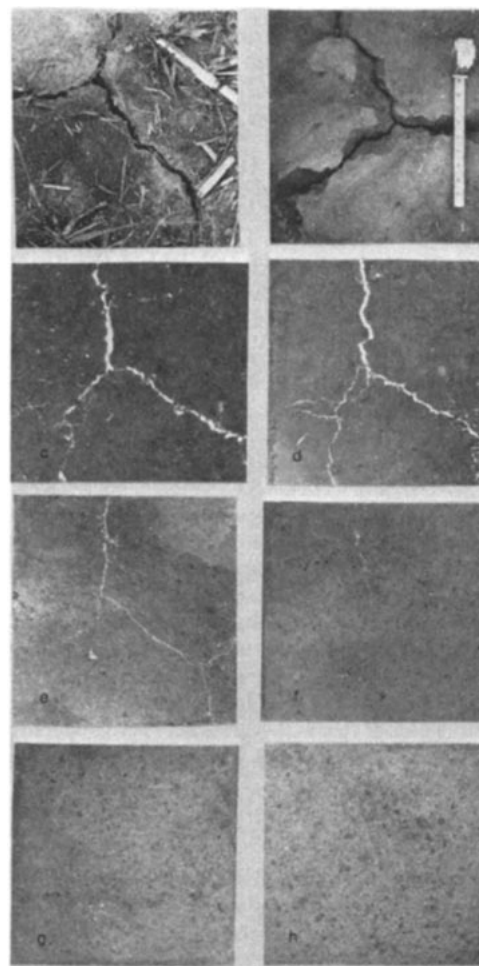


Fig. 1. Photographs showing cracks, root holes, worm holes, etc., filled with plaster of paris at different depths at no-till Site 2. (a) Soil surface before cleaning crop residue; (b) After cleaning, (c) At depth = 5 cm; (d) At depth = 10 cm; (e) At depth = 25 cm; (f) At depth = 35 cm; (g) At depth = 45 cm; (h) At depth = 55 cm.

posure than suggested by the exposure meter of the camera) had better contrast. These pictures were then analyzed on a Lemont Scientific OASYS (Optical Analysis SYStem, model DV-440) image analyzer for number, area, perimeter, and size distribution of macropores. This is a software-based system designed for analyzing optical images generated by a television camera with a resolution of 512 $\times$ 480 pixels and with external controls for adjusting the brightness and threshold grey levels. The system interfaces with a Digital Equipment Corporation (Northboro, MA) DEC LSI 11/73 microprocessor to produce an RS-170-format, TV-generated image by using an AP-512 Frame Grabber/Frame Buffer from Imaging Technology Incorporated, Woburn, MA. The system can partition the 256 grey levels of an image derived from a photograph, light microscope, or electron microscope into sixteen different pseudocolor bands. The system enables the operator to apply various digital image-enhancement and interactive-user techniques in order to obtain an accurate image.

The pores larger than 1600- μ m-equivalent diameter (visible) were considered as macropores. The pores larger than the minimum cutoff diameter were in the visibility range of the human eye, and they could be clearly distinguished by the image analyzer on the photograph of a horizontal soil plane. The pores having a diameter less than the minimum cutoff diameter could not be detected on the image analyzer

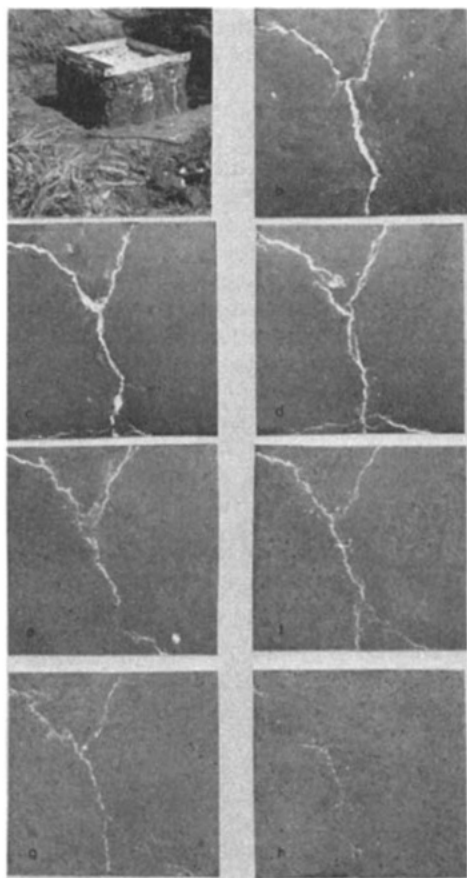


Fig. 2. Photographs showing cracks, root holes, worm holes, etc., filled with plaster of paris at different depths at conventional tillage Site 3. (a) Soil block impregnated with plaster of paris; (b) At depth = 5 cm; (c) At depth = 10 cm; (d) At depth = 15 cm; (e) At depth = 25 cm; (f) At depth = 35 cm; (g) At depth = 45 cm; (h) At depth = 55 cm.

because the pixel size limit was 1600 μm . Plaster-filled cracks and unfilled root or worm holes were analyzed by using two ranges of gray levels: 1 to 50 for dark, unfilled holes, and 160 to 255 for white, plaster-filled cracks. Since the soil under study was brownish yellow in color (moisture content 9–12%), the 1 to 50 grey level range was adequate to differentiate between dark unfilled holes and the background soil color. But in the black or nearly black colored soil, or in soil with high organic matter content it may be difficult to differentiate between unfilled holes and background. In that instance a different grey level range for the dark holes may have to be tried.

The total area analyzed at each plane was 2500 cm^2 . The total area, the perimeter, the number, and the size frequency distribution of macropores were determined for each depth (from the surface to a total depth of 60 cm in 5-cm increments). When calculating the number of macropores a continuous crack was treated as a single macropore. Figures 1 and 2 show the pictures of plaster-filled cracks and unfilled root, or worm holes at different depths under no-till and conventional tillage systems.

RESULTS AND DISCUSSION

Comparison of Methods

The plaster of paris technique was used to quantify macropores for no-till and conventional tillage plots. The main limitation of this method is that the plaster of paris does not infiltrate into pores less than about

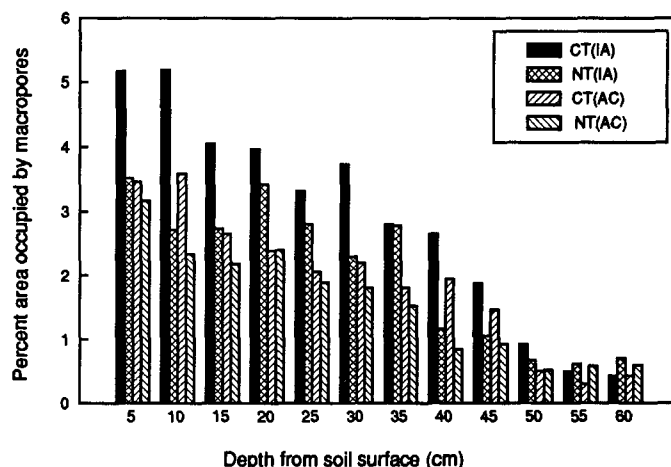


Fig. 3. Percentage area occupied by macropores as a function of depth (each bar represents average of three replications; CT = conventional tillage, NT = no-till; IA = automatic image analysis, and AC = AUTOCAD).

2 mm in diameter (FitzPatrick et al., 1985) or pores that are not connected to the soil surface. The pores not connected to the surface (or clogged at the surface) may be continuous onwards and contribute significantly to the preferential flow. These unfilled holes can easily be traced on transparent sheets, but on a photograph they appear as black spots, and it is difficult to analyze these dark holes and plaster-filled white cracks at the same time. This problem has not been addressed in previous studies. In this study, however, this problem was overcome by setting two different ranges of grey levels for white cracks and dark holes. This required the time-consuming effort of manually editing the images on the image analyzer to get accurate results.

Figure 3 shows the percentage area (average of three replications for each tillage treatment) occupied by macropores as a function of depth. The percentage area of macropores estimated by using the automatic image analysis was considerably greater than that estimated by the AUTOCAD method. Statistical analysis using a three-way factorial analysis of variance (three sources of variation, i.e., tillage system, method of measurement, and depth) showed that this difference was significant at the 99% confidence interval (the same confidence interval was used for other comparisons). There could be many reasons for this difference. One reason was the loss of information at two different stages. In the first stage, the image of a soil plane containing macropores is transferred onto a photographic film, and minute details can be lost when pictures are taken. For example, a narrow crack with a wavy outline could seem slightly thicker because of the lack of sharpness in the picture. Sometimes an unfilled portion within a crack (which is easy to trace on transparent sheet) can't be detected in the acquired image of the photograph because of similar problems. Some information can be lost in the second stage when the image is transferred from a photograph to the television camera connected to the image analyzer. At this stage, the information losses were based on the quality of the acquired image, which depended on many factors such as the intensity and angle of incident light

and adjustment of grey level of the pixels (especially when two different ranges of grey levels were used at the same time to pick up unfilled holes along with plaster-filled cracks). A compromise has to be made by the operator while deciding the threshold grey level because one category of features (e.g., white cracks) could be included more accurately only at the expense of losing another feature (e.g., dark holes). Oren (1987) found that an additional source of illumination inconsistency was caused by television camera lens itself. Although a live black-and-white display of images showed no apparent discrepancies to the naked eye, a pseudocolor analysis showed that the center portion of the image had a higher brightness intensity at all lens apertures. There was a variation of approximately 75 grey levels from the center of the field to the image edge. This could have caused overestimation of the area of the cracks in the central portion of the image. Greater variability within the replications in automatic image analysis in comparison to the AUTOCAD method is indicative of similar problems.

When macropores were traced on the transparent sheets, information losses were lower. External factors did not control the quality of the image and, therefore, the accuracy of measurements. To trace macropores on a transparent sheet, impregnation with plaster of paris is not required because the human eye usually can differentiate between a macropore and the soil matrix. The main limitation of this method is that it is time consuming. Tracing macropores on transparent sheets and digitizing them on AUTOCAD takes substantially longer (roughly 4–6 times) than image analysis. Although some error is involved in mapping the macropores, it is particularly difficult to trace small holes (which are sometimes just dots from the pen) on AUTOCAD.

Comparison of Tillage Treatments

Figure 3 shows that the total percentage area occupied by macropores decreased with depth under both tillage systems. The biggest decrease in percent area with depth was shown by image analyzing technique for the conventional tillage treatment, where the area decreased from 5% at 5 cm to 0.5% at 60 cm. On the average, the no-till sites had less macropore area in comparison to the conventional tillage sites. The difference was significant at the 99% confidence interval. This fact was qualitatively observed in the field while pouring plaster of paris on the selected sites. The sites in the conventional tillage plot consumed about 10 times more plaster of paris mixture than did the no-till sites. The reason for greater consumption of plaster of paris was that the sites in the conventional tillage plot had larger cracks in comparison to no-till sites. These cracks were not only larger but also deeper than those in no-till area. Visual observation of the photographs (Fig. 1 and 2) confirms this.

Figure 4 shows the change in total average macropore perimeter with depth under the two tillage treatments as measured by automatic image analysis and AUTOCAD. Both methods showed that the total average perimeter was variable (but not showing a definite increasing or decreasing trend) to a depth of 30 cm and then decreased at greater depth. The statistical

analysis (similar to that done for percentage area) revealed a significant effect of depth under both the tillage systems. Below a depth of 30 cm the size of cracks started to decrease significantly, causing the perimeter to decrease with greater depth (Fig. 1 and 2).

With both methods of measurement, the effect of tillage treatment on the total perimeter of macropores was statistically significant. Relatively greater number of worm holes or root holes in the no-till plot accounted for significantly greater values of macropore perimeter in comparison with the conventional tillage plot. Although the macropore-perimeter values estimated by automatic image analysis were not significantly different from those estimated by AUTOCAD, automatic image analysis gave smaller values of macropore-perimeter.

Figure 5 shows the total number of macropores as a function of depth. In general, the total number of macropores increased with increasing depth in both the tillage systems. The effect of depth was not statistically significant under the no-till system because of high variability among the replications, but was sig-

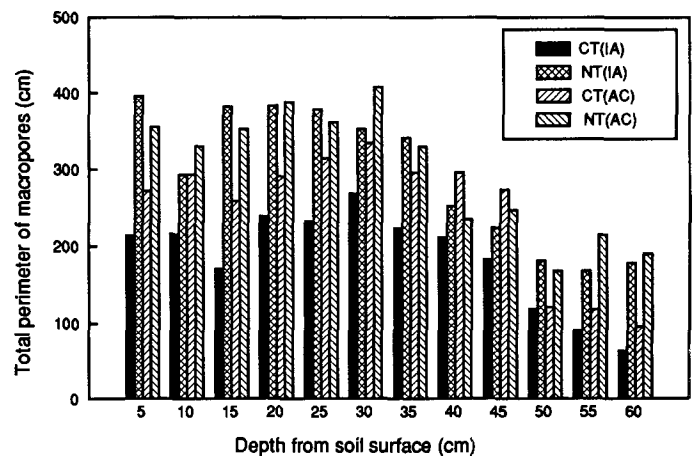


Fig. 4. Macropore perimeter as a function of depth (each bar represents average of three replications; CT = conventional tillage, NT = no-till, IA = automatic image analysis, AC = AUTOCAD).

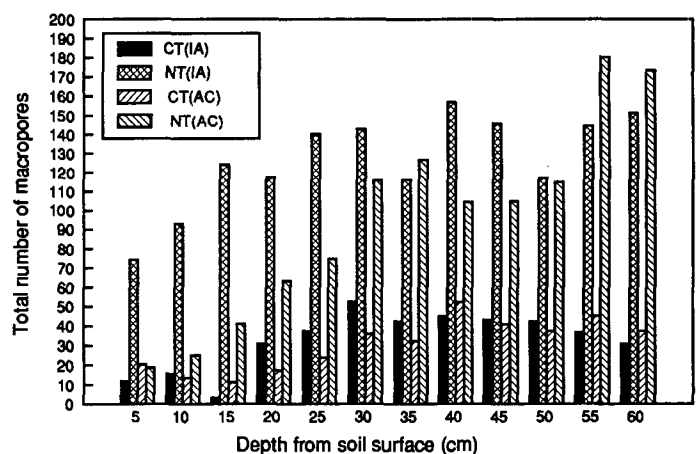


Fig. 5. Total number of macropores at different depths (each bar represents average of three replications; CT = conventional tillage, NT = no-till, IA = automatic image analysis, AC = AUTOCAD).

nificant under conventional tillage system. The total number of macropores was significantly greater for the no-till system than for the conventional tillage system because of more root holes, worm holes, ant holes, etc., in the no-till area, especially below 30 cm (Fig. 1). Although automatic image analysis estimated a relatively greater number of macropores than estimated by AUTOCAD, the difference was not statistically significant.

Size frequency distribution for the macropores was determined at each depth under both tillage systems by using the two methods of measurement. Figures 6 and 7 show the representative size frequency distributions at 15 cm (representing the upper 30-cm-soil profile) and at 45 cm (representing the lower 30-cm-soil profile), respectively, under both tillage systems. Size frequency distributions again indicated more macropores in the smaller size ranges (especially in the lower 30 cm of the soil profile) in the no-till area in comparison with the conventional tillage area. This indicates an abundance of root holes, worm holes, etc., in the no-till area.

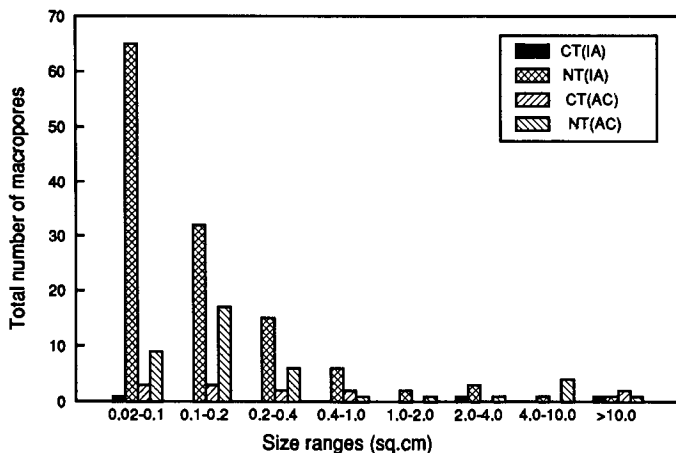


Fig. 6. Size frequency distribution of macropores at 15 cm (each bar represents average of three replications; CT = conventional tillage, NT = no-till, IA = automatic image analysis, AC = AUTOCAD).

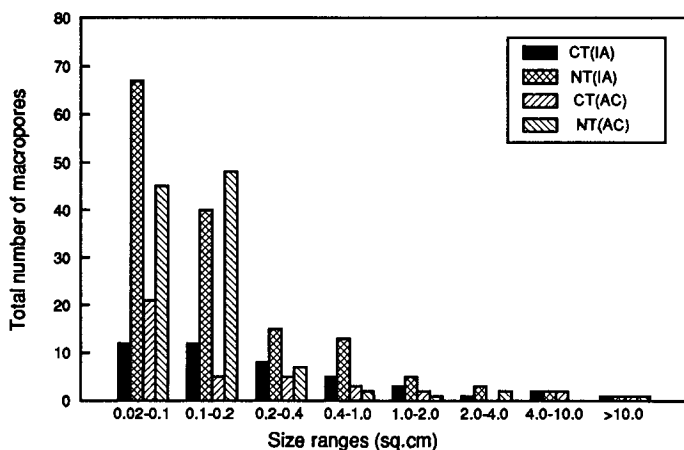


Fig. 7. Size frequency distribution of macropores at 45 cm (each bar represents average of three replications; CT = conventional tillage, NT = no-till, IA = automatic image analysis, AC = AUTOCAD).

CONCLUSIONS

The methods outlined here measure macropores of equivalent diameter greater than 1.6 mm but smaller macropores (less than 1.6 mm in diameter) could also be important in the flow process. For example, Watson and Luxmoore (1986) reported that 73% of saturated flux occurred in the pores of equivalent diameter greater than 1.0 mm. Further improvements can be made in these methods so that smaller macropores can be measured. Using an image analyzer with a better resolution or using an extra-sharp marker for the mapping of macropores may help in this direction.

Some specific conclusions from this study are as follows:

1. Tracing macropores on a transparent sheet and digitizing them on AUTOCAD proved to be a simpler and better (though more time-consuming) method to quantify macropores than automatic image analysis because fewer external factors were involved. Even the impregnation of a soil block with plaster of paris was not absolutely necessary in case of the AUTOCAD method.
2. The plaster of paris method has a limitation that small pores and the pores not connected to the surface can't be impregnated with plaster slurry. Those unfilled holes can't be detected by setting a single grey level range while automatically analyzing photographs of horizontal soil planes. This problem was solved by setting two different grey level ranges at the same time—one for dark, unfilled holes and the other for white, plaster-filled cracks.
3. The total percentage area occupied by macropores was greater for the conventional tillage system than for the no-till system because there were wider and deeper cracks under the conventional tillage system. The area of macropores decreased with greater depths under both tillage systems.
4. The total average perimeter of macropores was greater for no-till sites because of the greater number of macropores in the smaller size ranges. A larger number of smaller holes (especially below 30 cm) contributed less to the area and more to the perimeter. Under both tillage systems the total macropore perimeter decreased from 30 to 60 cm.
5. The total number of macropores generally increased with depth under both the tillage systems, indicating an increasing number of small worm holes, ant holes, and root holes. In the conventional tillage treatment, the total number of macropores was significantly less than the no-till treatment. Size frequency distribution of macropores also indicated a greater number of macropores in smaller size ranges for the no-till area (especially below 30 cm) in comparison to the conventional tillage area, indicating an abundance of root holes, worm holes, ant holes, etc., in the no-till area.

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